

3. A.

P_e = fraction of actual errors

P_m = fraction of syndrome measurement errors

we need to establish that:

highest possible energy of a correct solution $<$ lowest possible energy of a bad solution

decoder is guaranteed to work.

$$E(z) = \log \frac{1-P_e}{P_e} (\# \text{ bit flips}) + \log \frac{1-P_m}{P_m} (\# \text{ syndrome mismatches})$$

$$E(z) = \log \left(\frac{1-P_e}{P_e} \right) \cdot P_e \cdot n + \log \left(\frac{1-P_m}{P_m} \right) \cdot P_m \cdot n$$

energy for a correct solution

Now, for an incorrect solution $\hat{z}$

$$\frac{1}{n} \sum z_i \hat{z}_i \leq 0$$

which means that z_i and $\hat{z}_i$ disagree on at least half the bits: $\#(z_i \neq \hat{z}_i) \geq \frac{n}{2}$

So, originally we have $p_e n$ bit flips (all of these flips the decoder can disagree on)

+
we have $\left(\frac{n}{2} - p_e n\right)$ bits where the decoder
can disagree this is the lowest no. of bit flips
a decoder can register for the best bad
guess

$$E'(z) = n \left(\frac{1}{2} - p_e \right) \cdot \log \left(\frac{1 - p_e}{p_e} \right) + 0$$

for it to be the best bad guess

Now, $E(z) < E'(z)$

$$\log\left(\frac{1-p_e}{p_e}\right) \cdot n \cdot p_e + \log\left(\frac{1-p_m}{p_m}\right) \cdot n \cdot p_m$$

$$< n\left(\frac{1}{2} - p_e\right) \cdot \log\left(\frac{1-p_e}{p_e}\right)$$

$$\Rightarrow \left(\frac{1}{2} - 2p_e\right) \log\left(\frac{1-p_e}{p_e}\right) > \underbrace{p_m \log\left(\frac{1-p_m}{p_m}\right)}_{\substack{\text{this is small} \\ \text{if } p_m \text{ is small}}}$$

↑ this is $\leq \frac{1}{2}$ & this is large if p_e is small
↑

when $p_e = p_m = p$

$$\left(\frac{1}{2} - 2p\right) \log\left(\frac{1-p}{p}\right) > p \log\left(\frac{1-p}{p}\right)$$

$$\Rightarrow \frac{1}{2} - 2p > p$$

$$\Rightarrow 3p < \frac{1}{2}$$

$$\Rightarrow \boxed{p < \frac{1}{6}}$$

the question suggests $\leq \frac{1}{6}$ but there is no equality.